SOLARIS: SMART ENERGY MONITORING FOR SOLAR HOMES

Understand today's energy. Prepare for tomorrow's bill.

TEAM VISIONARIES

visionaries

THE CATCH

You invested in solar to lower your electricity bill.

SO WHY IS THE BILL STILL A SURPRISE?

OWNING THE EQUIPMENT DOES NOT MEAN UNDERSTANDING THE ENERGY

01 Customers cannot easily connect their daily energy use to their monthly bill.

02 Smart meters and hybrid inverters often display technical information.

03 Solar production, battery use, and grid consumption may appear in separate systems.

04 Customers discover problems only after receiving a high bill.

05 A new appliance or an always-on-load can raise the bill for weeks before anyone notices.

06 Predicting future consumption and expenses remains difficult.

SOLARIS EXPLAINS WHERE THE ENERGY AND MONEY GO

A smart energy monitoring device and mobile application for solar-powered homes.

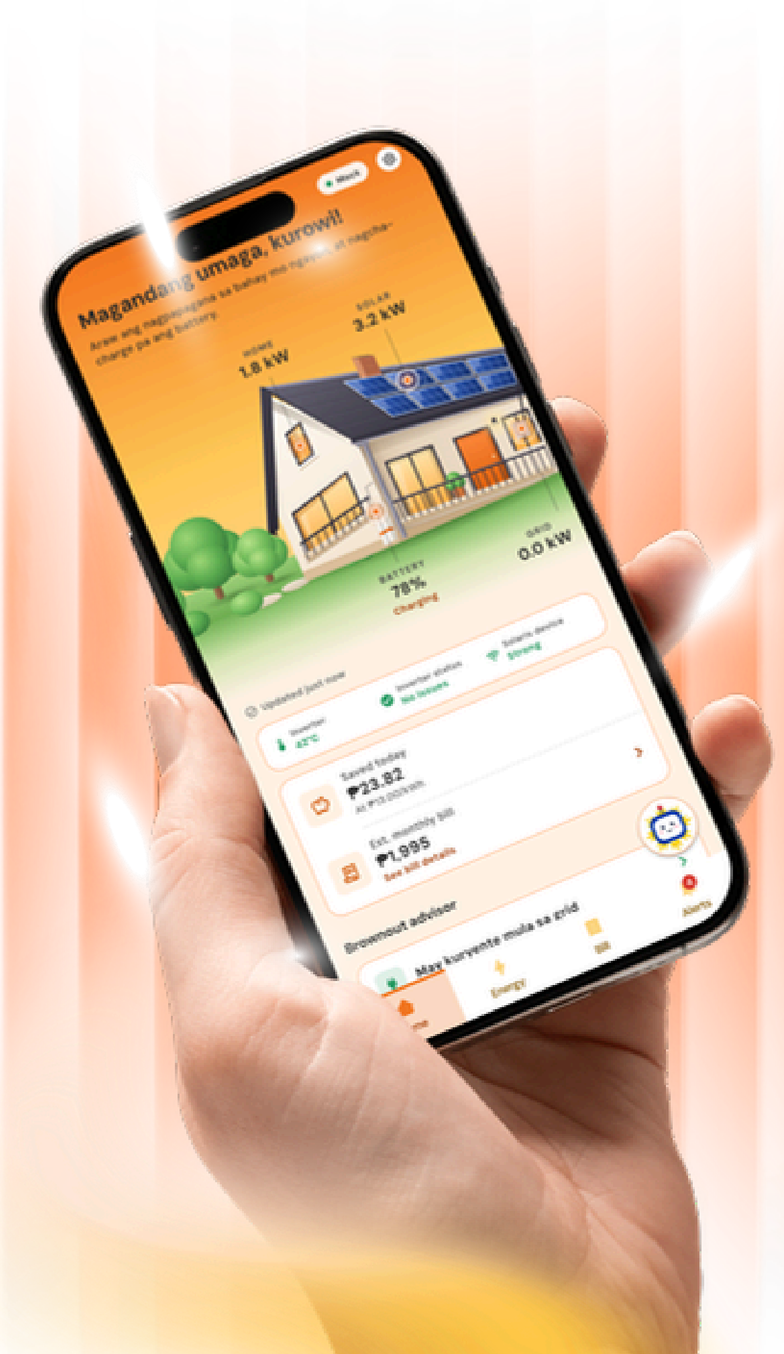
What the device reads:

- Solar Production
- Household Consumption
- Grid Usage
- Battery Status
- Inverter Condition

SOLAR
Hybrid Inverter
Solar + battery homes

METER
Smart Meter
Non-solar homes

BOTH
Inverter + Meter
Solar + exact billed kWh



FROM THE ENERGY SYSTEM TO THE CUSTOMER'S PHONE

01 • IN THE HOME

Smart meter or hybrid inverter

The equipment the home already has

02 • DEVICE

Solaris monitoring device

Reads the meter and inverter directly

03 • SERVER

Local server and prediction engine

Stores history, detects events, calculates costs, forecasts

04 • APP

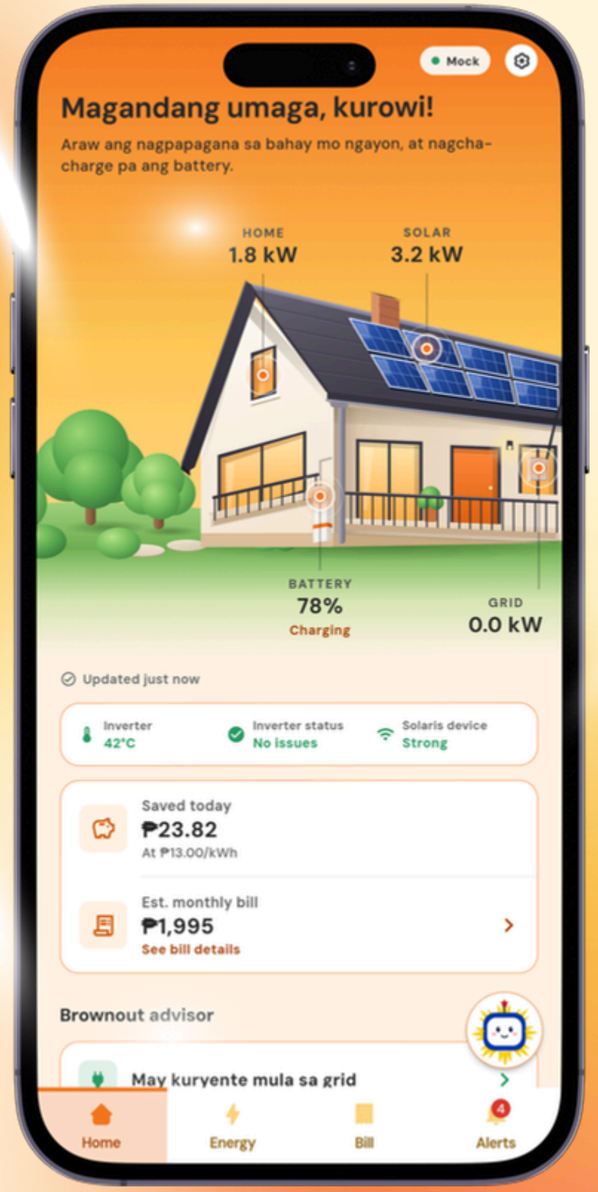
Solaris mobile application

Turns the data into information customers understand

05 • ASSISTANT

Mira AI energy assistant

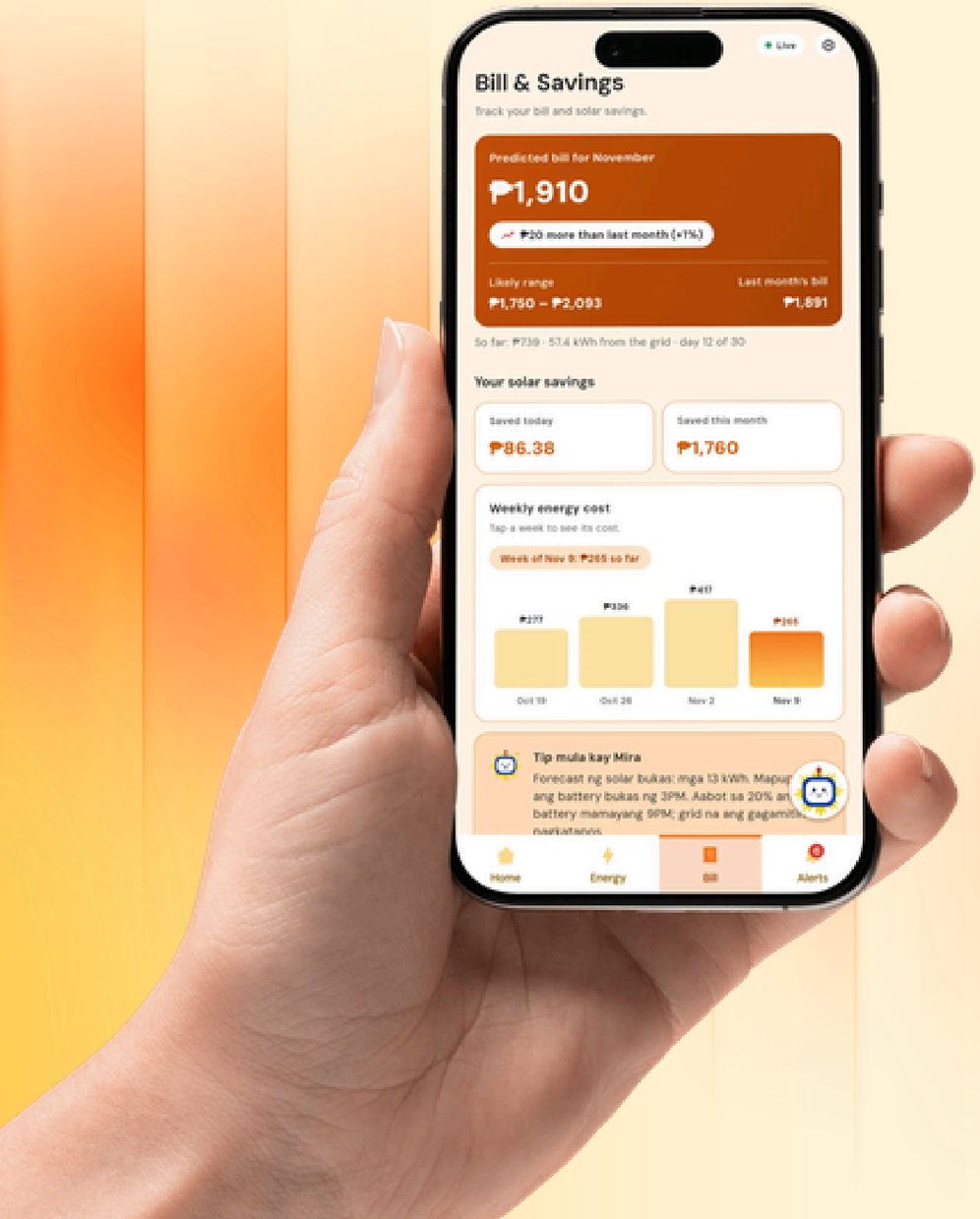
Explains the numbers and what to do next



WHAT IS HAPPENING AT HOME RIGHT NOW

- Current solar production
- Household electricity consumption
- Grid power, clearly labelled as importing or exporting
- Battery percentage and charging status
- Estimated savings for the day
- Inverter temperature and fault status
- Device connection and Wi-Fi status
- Brownout advisor: how long the battery would last if the power went out now

Updates live: switch on an appliance and see its effect within seconds.

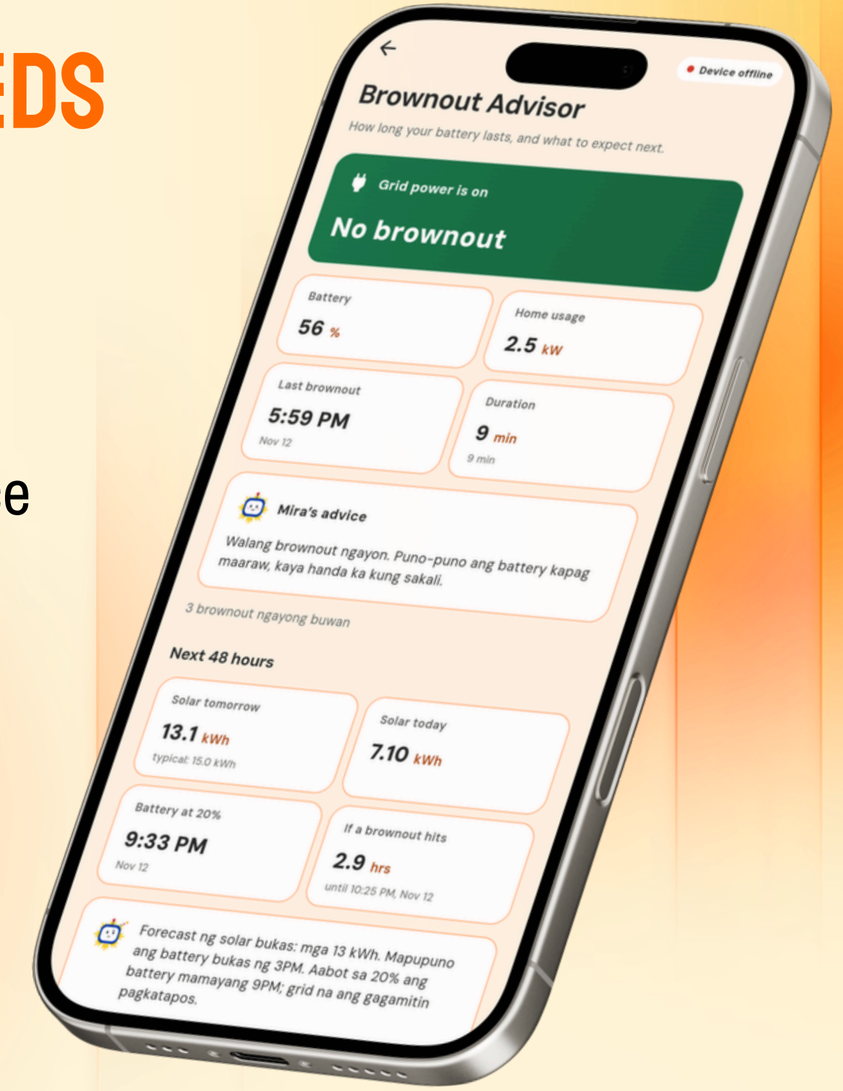


THE ELECTRICITY BILL BECOMES PREDICTABLE

- Daily, weekly, and monthly consumption
- Solar generation and grid usage
- Current month-to-date cost, priced with Meralco’s residential rate levels
- Predicted bill at the end of the month, with a low and high range
- Coming months: up to three months ahead, clearly marked as estimates
- Comparison with the previous month
- Estimated solar savings

PLANNING FOR TOMORROW'S ENERGY NEEDS

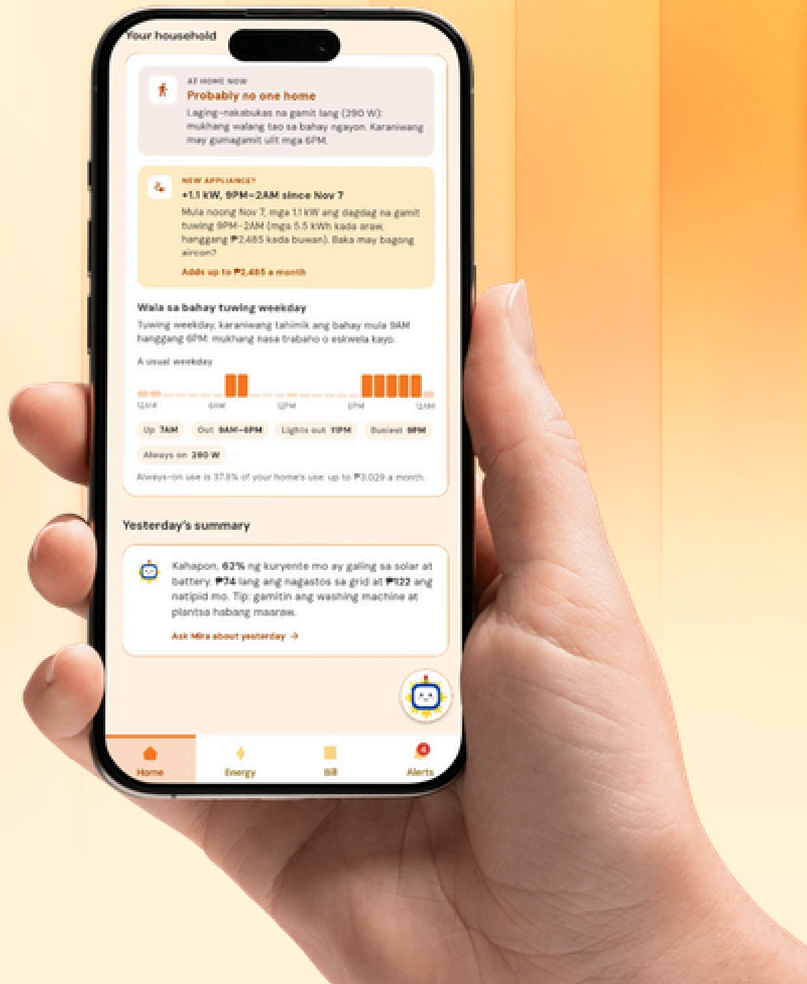
- Expected solar production for the next 48 hours
- Weather-informed forecast using hourly solar radiation and cloud conditions
- Forecast calibrated with the home's actual solar performance
- Projected battery percentage
- Expected grid consumption
- Estimated time until the battery reaches its reserve
- Estimated battery runtime during a brownout, including the expected sun
- History-based forecast when weather data is unavailable



GETTING MORE FROM THE SUN THE HOUSEHOLD ALREADY HAS

Smart recommendations to maximize solar energy, reduce electricity costs, and improve system efficiency.

- **Best Solar Window**, identifies the three sunniest hours with the most available solar energy.
- **Smart Solar Tips,** real-time appliance recommendations based on solar availability
- **Energy Habits,** personalized tips based on the last 30 days, ranked by potential saving
- **Clean Energy Impact,** tracks clean-energy usage CO2 emissions avoided, and equivalent trees
- **System Check,** evaluates solar, battery, and inverter performance and suggests upgrades with estimated savings and payback periods



SOLARIS LEARNS THE HOME'S ROUTINE FROM ITS ELECTRICITY USE ALONE

- **Household Profile:** Learns when occupants are usually home or away.
- **Daily Routine:** Identifies usual wake-up, sleep, and peak activity hours.
- **Live Household Status:** Estimates if someone is home, asleep, or away.
- **Unusual Activity:** Detects unexpected electricity usage.
- **Always-On Load:** Tracks standby consumption and monthly cost.
- **New Appliance Detection:** Identifies possible newly added appliances.

SOLARIS FOR HOMES WITHOUT SOLAR

- **Live Meter Reading:** Monitors power, voltage, and current.
- **Voltage Check:** Detects abnormal voltage levels.
- **Usage Patterns:** Tracks consumption and peak hours.
- **Meralco Rate Tracker:** Monitors monthly consumption levels.
- **Cost Alerts:** Warns users about potential bill increases.
- **Always-On Load:** Tracks standby usage and costs.
- **Brownout Alerts:** Notifies users of outages and restoration.
- **Mira AI:** Answers questions about electricity and savings.

MIRA EXPLAINS THE CUSTOMER'S ENERGY DATA

In simple Taglish, using the household's actual readings.

"Gaano katagal ang battery?"

"Paano ko mapapababa ang bill this month?"

"Kailan pinakamataas ang consumption namin?"

"Okay ba ang solar production ko?"

CLEAR GUIDANCE WHEN SOMETHING CHANGES

Brownout Alert

With the estimated battery runtime at the current load

Low-Battery Warning

Before the battery reaches its reserve

High-Temperature Warning

When the inverter hotter than usual

Inverter Fault Explanation

What the fault means, in plain words

Device Offline Notification

When Solaris stops receiving readings

Recommended Actions

A practical response for current conditions

A BETTER-INFORMED CUSTOMER IS AN EASIER CUSTOMER TO SERVE

Available Today

- Bill forecasting and rate warnings
- Mira AI customer assistance
- Household usage tracking
- Brownout and voltage recording

With Meralco Partnership

- Opt-in customer data sharing
- Direct outage reporting
- Area-wide outage awareness
- Meralco-led energy-saving programs

A BETTER-INFORMED CUSTOMER IS AN EASIER CUSTOMER TO SERVE

Current Customer Experience	With Solaris
The cost is discovered when the bill arrives	The bill is forecast before the month ends
The inverter displays technical readings	The app provides clear household explanations
Problems may be noticed late	Live alerts identify faults and brownouts
Battery runtime is uncertain	Remaining runtime is estimated
Solar savings are unclear	Daily and monthly savings are visible
Information is spread across different devices	One application brings the information together

WHY SOLARIS STANDS OUT?

ENERGY FOR EVERY HOME

Solar or meter-only, Solaris works for everyone.

FORECASTS YOU CAN TRUST

Tested predictions with honest estimates and ranges.

AI THAT SHOWS ITS SOURCES

Mira gives answers backed by actual home readings.

YOUR DATA STAYS YOURS

Private, processed inside your home, even offline

MADE FOR FILIPINO HOMES

Meralco rates, brownout advice, and Taglish support.

SOLARIS

SOLARIS HELPS SOLAR OWNERS UNDERSTAND THEIR ENERGY TODAY AND PREPARE FOR ITS COST TOMORROW.

Visibility · Preparation · Control over household energy costs